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Introduction

The United States is one of the most complex and wide countries, hosting a staggering population of 331,449,281 according to the 2020 Decennial Census [1] with very different backgrounds, making this country a significant actor on the global stage. Understanding how this country and its population are going to evolve in the close future is fundamental to make informed decisions in a variety of fields, including health, insurance, pension system and economic development.

Objective

The main scope of this research is to investigate how the amount and structure of the American population changed over the past century, and how this was caused by changing trends in mortality and fertility. To gain insights of the factors that have shaped population dynamics, both the life expectancy at birth and the total fertility rate will be studied to understand how main historical events caused changes in their intensities. Such

events include the end of the Second World War in 1945, with the following economic and demographic expansion, the improvement in the access to healthcare gained with the institution of Medicare and Medicaid in 1965, along with the advancements in medical technology reached in the 1980s, with the development of Computed Tomography (CT) scans and Magnetic Resonance Imaging (MRI) machines, prosecuted in the following decades with the vaccines development and the discovery of and Antiretroviral Therapy (ART) for HIV. Lastly, the recent effect of the COVID-19 pandemics will also be explored.

By studying the historical trends and development of mortality and fertility patterns in the United States, it is possible to forecast these trends for the next decades making assumptions on the underlying mortality model. In this way it is possible to predict the evolution of these demographic indicators over the next 50 years, in order to get insights about future demographic shifts, which can be valuable in various fields.

Relevance

As stated above, understanding past and future trends of mortality and fertility can be valuable for several areas, such as insurance, health, and pension. Regarding the insurance sector, a core information needed to design appropriate risk models and determine premiums is the evolution of mortality over the years. Also, understanding mortality and fertility patterns is crucial to develop retirement plans and pension schemes accordingly to the evolution of the population age structure, which in the case of the United States is represented by an aging population. Lastly, in the healthcare domain, these demographic dynamics are essential to keep providing quality services in terms of resource allocations and healthcare planning. However, demographic analysis is crucial also for economic coordination, which encompasses labour market, social security programs, and economic growth projections.

To get more insights of the underlying differences present within this country, the analysis will also explore gender differences in mortality, to shed light on disparities in health outcomes and cultural causes. Indeed, identifying gender-specific patterns is valuable to policymakers to develop targeted policies that try to address inequalities and foster the overall wellbeing of the population.

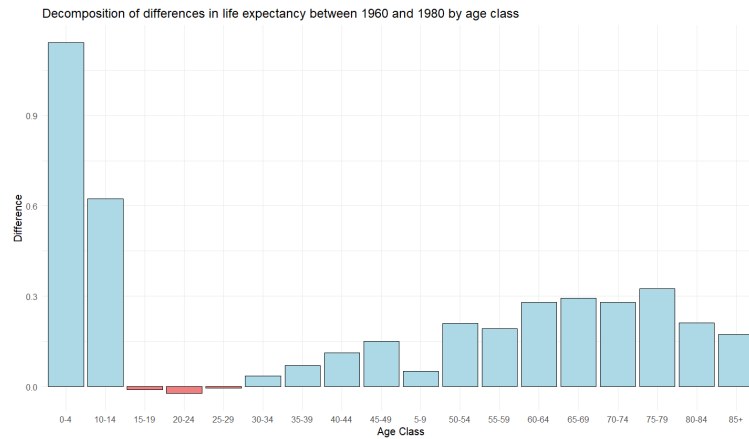
Socio-Economic Characteristics of the United States

Before analysing the trends of fertility and mortality over years, it is convenient to provide an overview of the main socio-economic characteristics of the United States. As stated above, it is indeed a highly polyhedric country, with a variety of factors that influence its demographic development [2]. In particular, this chapter provides an abstract of healthcare system, GDP per capita, employment rates and education attainment, keeping in mind that complex phenomena such as mortality and fertility are highly influenced also by individual lifestyle and cultural dynamics. Lastly, a cross-sectional view of the distribution of the population by age will be given and its transformation over the past century will be explored to notice the changes in the proportion of young, adult, and elderly populations.

Healthcare System

The American healthcare system is a complex system that combines public and private funding. During the past century it has undergone many transformations [3], including the institution of two government-funded healthcare programs in 1965, namely Medicare, designed for individuals older than 65, and Medicaid, which offers coverage to low-income individuals. This benchmark in the healthcare access contributed to an increase in the life expectancy of the elderly population as by providing affordable access to healthcare, Medicare speeded up the detection and managing of chronic diseases [4]. Indeed, as shown in Figure 1, the contribution to the difference in life expectancy between 1960 and 1980 is quite high in the individuals aged more than 60 than in the young and adult population.

Figure 1. Decomposition of differences in life expectancy by age between 1960 and 1970.



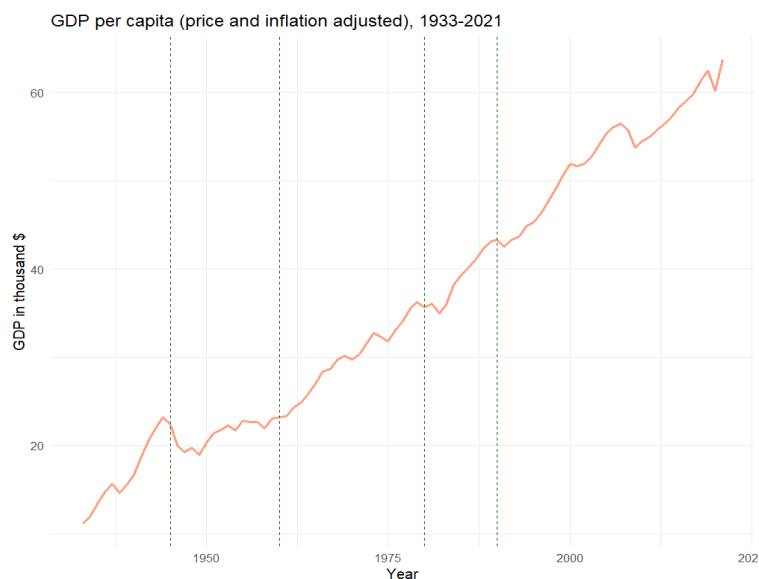
However, both Medicare and Medicaid did not show a direct effect on the total fertility rate [5], since the American healthcare system still perpetuates disparities in the access to medical services, which can influence reproductive choices as well as mortality patterns.

Socio-Economic Indicators

Due to the huge disparities present in the United States, it is interesting to analyse how some socio-economic indicators, such as the income and the educational attainment evolved in relationship with demographic trends. Indeed, the well-being of a population has a extreme impact both on fertility levels and life expectancy.

Firstly, the Gross Domestic Product (GDP) growth rate depicted in Figure 2, which is normally used as an indicator of economic expansions and contractions over time, was used to explore the association with increased or poorer opportunities for prosperity and access to healthcare which both impact the general wellbeing of an individual and its aspirations to create a family.

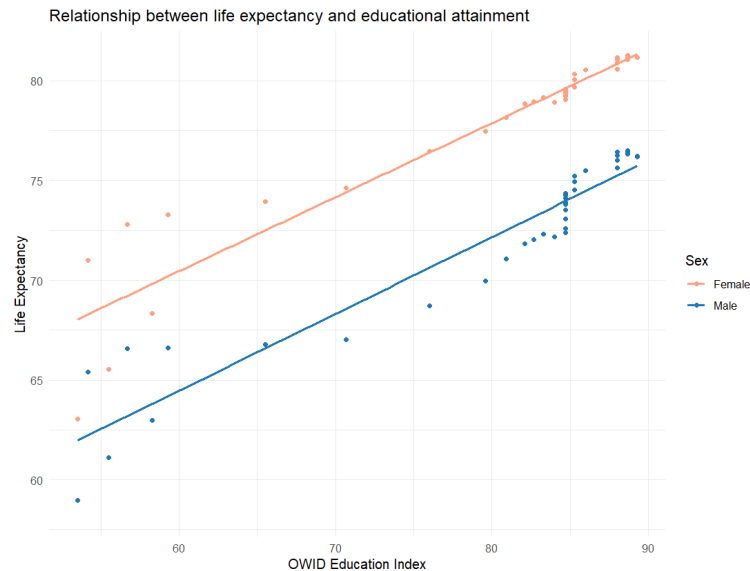
Figure 2. Evolution of GDP in the US from 1933 to 2021.



Beside the GDP, also employment rates reflect the situation of job opportunities within a country, and they are highly determined not only by the economic situation but also by the educational attainment. Indeed, an higher education level is a significant determinant both for health outcomes and fertility rates. In particular, a higher education provides access to more employment opportunities with higher income, and also is related

to better health knowledge [6]. Both factors contribute to improved health behaviours both in terms of prevention and treatment, and this relationship can be seen in Figure 3, where the life expectancy is related to the OWID (Our World In Data) Education index, which measures educational attainment through literacy and enrolment rates in primary, secondary, and tertiary education, in addition to the number of completed years of school. The positive relationship seems to be similar among males and females, with an almost constant difference in terms of life expectancy between the two sexes.

Figure 3. Relationship between educational attainment and life expectancy in the US from 1933 to 2021.



Moreover, educational attainment also influences fertility decisions, as individuals with higher education level are facilitated to find a highly paid job position, which increases both the well-being of the family and the individual satisfaction, making more desirable the option to have children [7]. However, as for all complex phenomena, it is crucial to consider also individual aspirations and cultural dynamics as they can change the theoretical relationship between these factors. Indeed, the cultural shifts led to a decrease in fertility rates as individuals tend to prioritize their career goals instead of making family targets.

Cultural Dynamics and Individual Choices

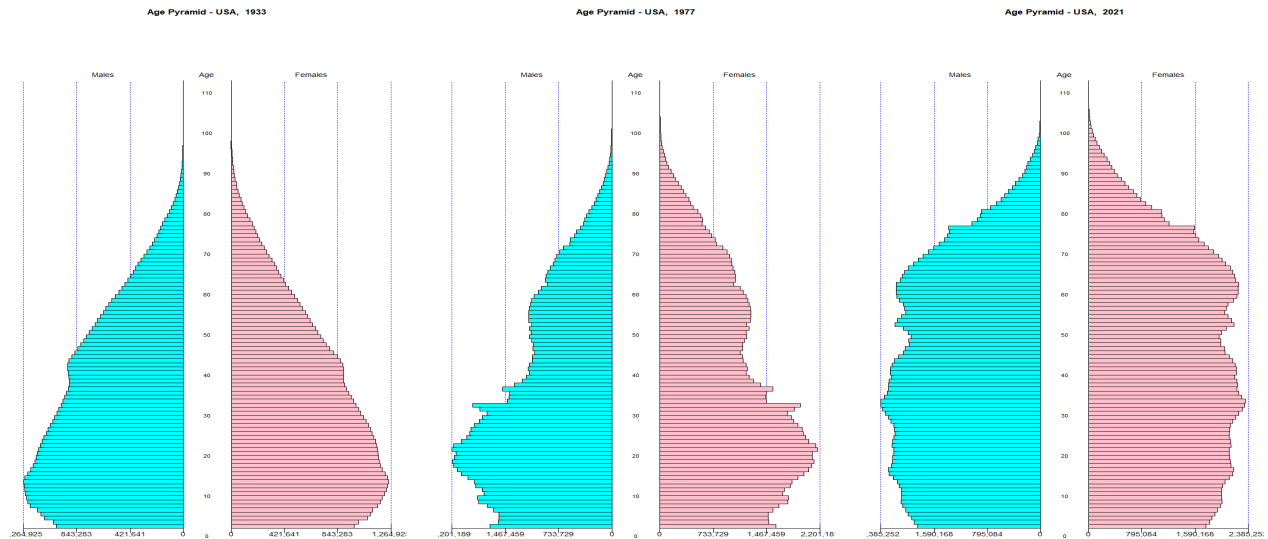
Both mortality and fertility patterns are highly influenced by individual choices and cultural dynamics, especially in such a complex country as the United States. In the 1980s, the United States made the first step towards family planning with an increase in the use of contraceptives. In that period family planning started to be accepted as a means to reach the desired family size [8]. As a consequence, the fertility rates declined, reflecting the desire to manage family size and to prioritize individual aspirations [9]. Then in the following decade, societal norms saw a massive change in terms of increased opportunities for women, which helped them to pursue a higher education and better jobs, with a general increase in their independence [10]. This led the size of the families to keep decreasing, since family planning decisions were in line with the evolving individual lifestyles.

It is important to highlight that cultural dynamics and individual choices are interconnected and influenced by common factors. For instance, economic stability, educational opportunities, and the reduction of the gender gap have all played significant roles in shaping both societal dynamics and individual decision-making process regarding healthcare and family dimension.

Population Structure

All the previously explained factors had an impact on the evolution of both fertility and mortality, which in turn defined the population size and structure of the United States, in terms of sex and age composition, as shown in the age pyramids in Figure 4.

Figure 4. Age pyramid of the US in 1933, 1977 and 2021.



Regarding fertility, one of the most influential events was the post-war baby boom between 1946 and 1964 which was characterized by a remarkable increase in birth rates, resulting in a bulge in the population pyramid in the younger age groups. The following improvement in the healthcare system which contributed to a higher life expectancy and a decrease in mortality rates among old people, resulted in a widening of the older age groups within the pyramid, since it made possible for a larger amount of people to reach advanced ages. This aging trend in the population was reinforced by the declining birth rates in recent decades.

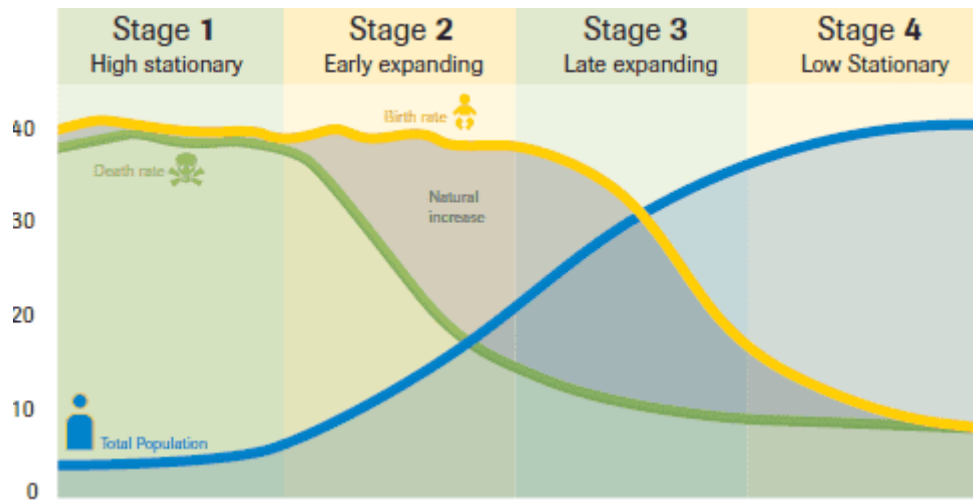
As a result, the United States has experienced a demographic transition characterized by a smaller proportion of young individuals relative to older age cohorts. These changes have important implications for various aspects of society, including healthcare, retirement planning, labour force dynamics, and social welfare programs. The healthcare sector faces the challenge of meeting the growing healthcare needs of an aging population, including managing chronic conditions, providing long-term care, and addressing age-related health concerns [11]. Also pension programs and social security systems are influenced by this demographic shift, since the aging population created sustainability issues, as a larger proportion of retired population relies on this financial support. Finally, this trend poses challenges in the labour market as younger working force will be compelled to work beyond traditional retirement ages in order to supply to the intergenerational dependency [12].

Theoretical Framework

The Demographic Transition Model

The Demographic Transition Model (DTM) is a framework that allows to understand how changes in fertility and mortality rates impacted population dynamics of a country over the years [13]. This theory identifies four distinct stages that reflect the changes in birth and death rates, resulting in shifts in population increase and structure. Figure 5 represents its theoretical structure with the subsequent stages.

Figure 5. The four stages of the Demographic Transition Model.



During the first stage, known as the pre-transition stage, both birth and death rates are high, leading to a relatively stable population size. Indeed, limited access to modern healthcare and sanitation facilities contributes to high mortality rates, which hinder population growth.

As societies progress to the second stage, the early transition stage, they experience a decline in death rates thanks to industrialization and advancements in healthcare. However, birth rates remain high, resulting in a substantial population growth. Thus, the age pyramid during this stage often showcases a broad base, indicating a significant proportion of young individuals benefiting from improved survival rates.

In the third stage, the late transition stage, birth rates start to decline due to social and economic changes, such as an increased access to education and evolving societal norms which promulgate lower fertility rates. As a result, the natural increase of the population slows down, and the age pyramid begins to shift, reflecting a more balanced distribution across different age groups.

Lastly, in the post-transition phase both birth and death rates reach low levels, approaching a near-replacement fertility rate, which is the average number of children per woman needed to maintain the population size stable over time. Therefore, the population is more evenly distributed across the different ages, so the pyramid takes on a more rectangular shape.

Over the past century, also the United States have undergone a significant demographic transition [14], with declining death rates and high birth rates in the early 20th century, which contributed to the increase of population size. However, after the baby boom period, the birth rates started to decrease due to access to contraception and changes in societal norms and general living conditions. This made the population pyramid to become more rectangular as it is nowadays. In fact, the United States are currently in the post-transition stage, with low rates of fertility and mortality, which are characteristics of developed and industrialized societies.

The Life Table

To conduct the analysis of demographic indicators, life tables have been constructed based on population data to study the probability of death of individuals over time using age-specific death rates and population size. The table is constructed starting from a fictive radix population of $l_0 = 100,000$ individuals. From this value, several statistics were derived as follows [15]:

- $l_x = \frac{d_x}{q_x}$ is the number of survivors at age x and represents the number of individuals who have survived up to age x in the given population.

- $q_x = \frac{d_x}{l_x}$ is the probability of dying between ages x and $x+1$ and represents the probability that an individual aged x will die before reaching the next birthday.
- $d_x = l_x q_x$ is the number of deaths between age x and $x+1$ and denotes the number of individuals who die between ages x and $x+1$ in the population.
- $L_x = l_{x+1} + 0.5 d_x$ is the number of person-years lived between age x and $x+1$, which represents the cumulative sum of the years lived by individuals who survive from age x to $x+1$.
- $T_x = \sum_{x=0}^{\omega} L_x$ is the total number of years lived by the cohort in age x and indicates the sum of person-years lived by the entire cohort from birth up to age x .
- $e_x = \frac{T_x}{l_x}$ is the life expectancy at age x and represents the average number of additional years a person aged x can expect to live.

Data and Sources

To get access to mortality rates and population size of the United States, the Human Mortality Database (HMD) was used, which provided data from 1933 to 2021 regarding mortality patterns in the United States [16]. In addition, the Human Fertility Database (HFD) was used to gain access to total fertility rates. Since it is aligned with the study period from 1933 to 2021 [17], the HFD allowed to examine fertility patterns and trends throughout this timeframe. By incorporating the two datasets, it was possible to explore the demographic aspects related to mortality and fertility rates in the United States.

In addition to demographic data, Gapminder was used to obtain data regarding several socio-economic characteristics such as GDP, educational attainment, and employment rates [18]. Incorporating Gapminder data into the analysis provided valuable insights into the socio-economic context of the United States which was fundamental to understand demographic trends. The time period covered by the Gapminder data aligns seamlessly with the study period, however it includes some missing data points due to differences in data collection methods.

Methods

The Lee-Carter Model

The Lee-Carter model is a widely used statistical model in demography for analysing mortality data. It was developed by Ronald Lee and Lawrence Carter in 1992 [19]. The model aims to capture and forecast age-specific mortality rates over time by decomposing the age-specific mortality rates into two components: a time-varying common pattern shared across all ages and a set of age-specific deviation terms. It is defined by the following equation:

$$\ln m_x(t) = \alpha_x + \beta_x K_t + \epsilon_{x,t}$$

Where $m_x(t)$ is the mortality rate at age x and time t , α_x captures the age-specific rate of mortality improvement, and k_t captures the time-specific common pattern in mortality rates, assuming that it follows a stochastic process.

This model can be fitted in R using the 'demography' package, which provides a linear approximation approach transforming the mortality rates into logarithmic form, fitting a linear regression model, and estimating the parameters α_x , β_x , and k_t using ordinary least squares. This approach is computationally efficient and easy to implement, however the linear relationship assumption between log-mortality rates and the time component may not hold in all cases. On the other hand, this model can also be estimated with the 'StMoMo' package which offers maximum likelihood estimation for the parameters, being more flexible and able to capture nonlinear patterns, even if computationally more intensive.

We estimated the Lee-Carter model to forecast life expectancy in the United States, employing mortality data spanning from 1933 to 2020, by applying both the linear approximation approach in the 'demography' package and the maximum likelihood estimation in the 'StMoMo' package. The parameters are the baseline mortality level (α_x), the rate of mortality improvement (β_x), and the time-specific common pattern in mortality rates (k_t). Then we assessed the model's performance by analysing residuals, which represent the differences between the observed and predicted mortality rates.

The Renshaw-Haberman Model

In addition to the Lee-Carter model, we also employed the Renshaw-Haberman model to analyse mortality data and forecast life expectancy. The Renshaw-Haberman model, introduced by Andrew Renshaw and Steven Haberman, is another popular approach used in demography as it expands upon the Lee-Carter framework by incorporating cohort effects. In this way it is able to capture the impact of historical events or birth cohorts on mortality rates. It is defined by the following equation [20]:

$$\ln m_x(t) = \alpha_x + \beta_x^{[0]} K_t + \beta_x^{[1]} \gamma_{t,x} + \epsilon_{x,t}$$

The Renshaw-Haberman model assumes that the logarithm of the age-specific mortality rates follows a parametric function of age and time. In this equation, α_x and $\beta_x^{[0]}$ are the same as in the Lee-Carter model, capturing the age-specific baseline mortality level and rate of mortality improvement. The additional terms $\gamma_{t,x}$ and $\beta_x^{[1]}$ represent the cohort-specific deviation from the baseline and the cohort-specific time-varying effect, respectively. So, by estimating the parameters of this model we can examine both period and cohort effects on mortality rates. Also for this model we assessed the goodness of fit by examining the residuals, which helped evaluate the model's performance and identify any systematic deviations between the observed and predicted mortality rates.

Lastly, to compare the Lee-Carter model and the Renshaw-Haberman model, we computed the Bayesian Information Criterion (BIC) values for the two fitted models with the following formula [21]:

$$BIC = -2 \cdot \log\text{likelihood} + k \cdot \log(n)$$

where log-likelihood represents the maximized value of the log-likelihood function, k represents the number of parameters in the model, and n represents the sample size. Thus, the model with the lower BIC is considered to have a better balance between goodness of fit and parsimony. The BIC criterion considers both the model's fit to the data and its complexity, ensuring that the chosen model achieved a balance between accuracy and complexity when analysing mortality data and forecasting life expectancy.

Smoothing Life Expectancy

In the analysis of life expectancy dynamics, we employed the Hodrick-Prescott and Baxter-King filters to smoothen the time-series data for the US dataset. These filters, widely used in business cycles, proved to be valuable tools in detrending the data and analysing cyclical fluctuations. By applying these filters specifically to life expectancy data, we gained insights into the long-term trends and cyclical variations in life expectancy over time. The Hodrick-Prescott filter [22], initially developed to study business cycles in the post-war US economy, effectively separated the trend component from the cyclical component of life expectancy data. This allowed for a focused analysis of the underlying patterns and dynamics. Additionally, the Baxter-King filter, building upon the Hodrick-Prescott filter, further refined the analysis by isolating the repetitive component of the time series. By specifying the width for the periodic oscillations, it enhanced the detection and interpretation of cyclic fluctuations in life expectancy. The application of these filtering techniques provided valuable insights into the trends and cyclical variations in life expectancy, contributing to a comprehensive analysis of life expectancy dynamics over time in the United States.

Results and Discussion

Demographic Trends

The United States experienced many significant historical events over the past century, which also contributed to shaping its demographic characteristics. Specifically, Figures 6 and 7 provide the evolution of mortality and fertility dynamics through the representation of life expectancy at birth (e_0) by sex and the total fertility rate (TFR) from 1933 to 2021.

Figure 6. Evolution of life expectancy in the United States from 1933 to 2021 by sex.

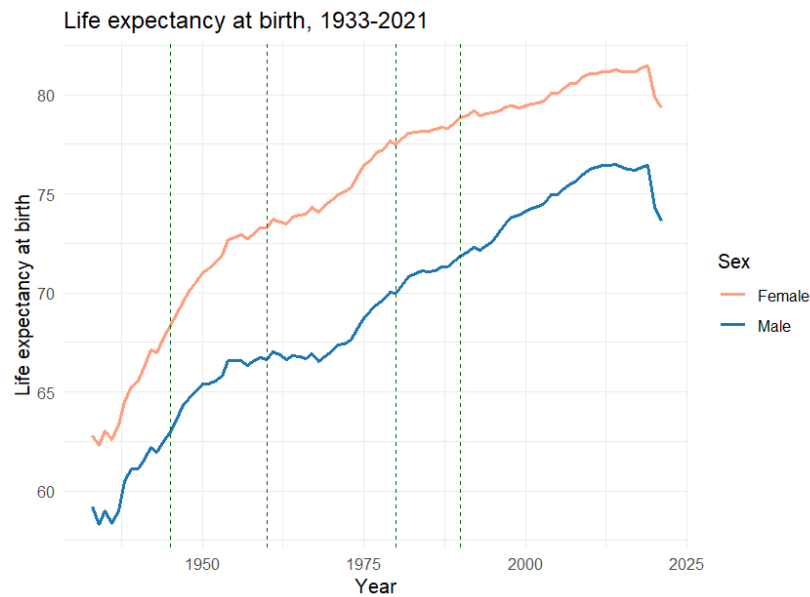
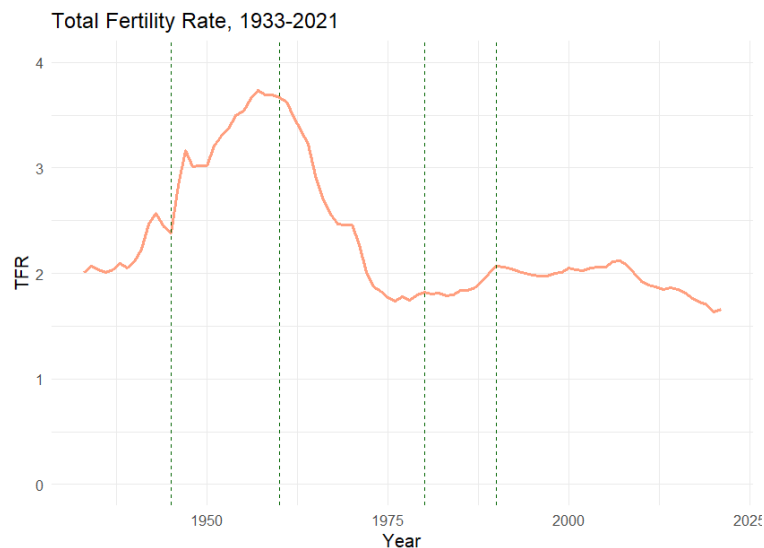


Figure 7. Evolution of total fertility rate in the United States from 1933 to 2021.



During the period from 1935 to 1945, the United States experienced many advancements in the medical and healthcare delivery systems. These developments contributed to modest improvements in life expectancy, among both males and females, however, as medical interventions and treatments became more effective, also life expectancy increased more rapidly. Despite the economic hardships of the Great Depression and the disruptions caused by World War II, the total fertility rate exhibited an unexpected slightly increasing trend during the same period.

The post-war era, particularly between 1945 and 1960, marked a significant turning point in the American demographic landscape. The economic expansion that followed World War II brought about improved living standards, increased job opportunities, and a general sense of optimism. These favourable conditions, coupled with a surge in marriages and a desire to rebuild families, led to the baby boom generation, which was a large cohort born between 1946 and 1964 resulted from the significant increase in fertility rates. The baby boomers not only contributed to population growth but also shaped social, economic, and cultural dynamics within the United States.

Life expectancy kept rising during this period and also in the following decades for both males and females, thanks to the contribution of advancements in healthcare, medical technology, and disease prevention. Additionally, increased public health awareness, access to healthcare services, and advancements in medical treatments further supported the extension of life expectancy.

In contrast to the increasing trend in life expectancy, the total fertility rate exhibited a notable decline in the decade that followed the baby boom, mainly due to the widespread availability and use of effective contraception methods, coupled with changing societal attitudes towards family planning. In addition, increasing educational and employment opportunities for women, as well as changing gender roles, contributed to delayed childbirth and a decrease in the desire for larger family sizes.

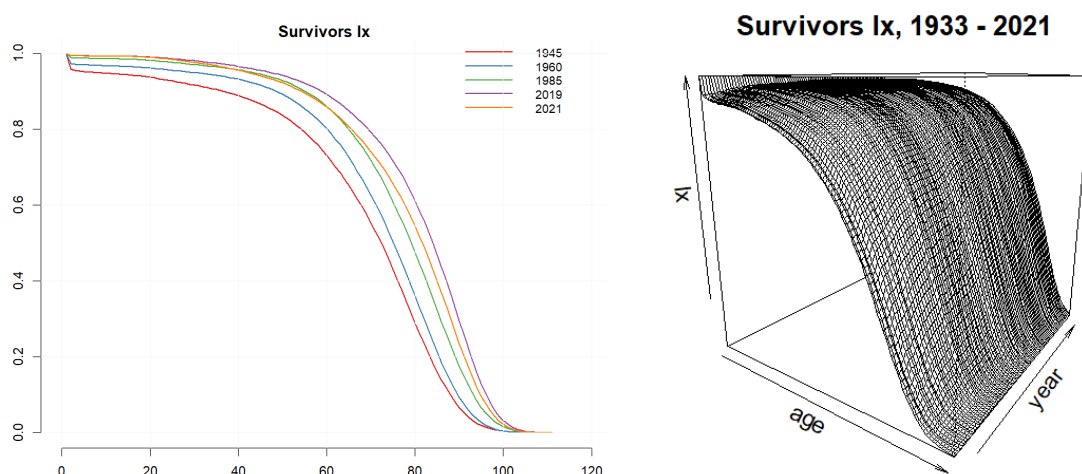
These trends in life expectancy and fertility rates are consistent with the later stages of the demographic transition model. As the United States progressed towards a more developed and modern society, characterized by advanced healthcare systems, urbanization, and changing societal values, life expectancy continued to increase while fertility rates continued to decline, remaining below the replacement threshold of 2.1 births per woman.

Life Table Measures

To explore demographic dynamics of the American population, the life table was constructed both for the whole population and divided by sex. It allowed to examine the changes in the number of survivors and deaths at different ages, and to compute measures such as life expectancy.

The number of survivors (l_x) represents the individuals who have survived to a specific age and provides a measure of the likelihood of survival at different stages of life. Figure 8 illustrates the evolution of the number of survivors in the United States life table from 1933 to 2021, both in a bidimensional and three dimensional perspective.

Figure 8. Evolution of the number of survivors from 1933 to 2021.

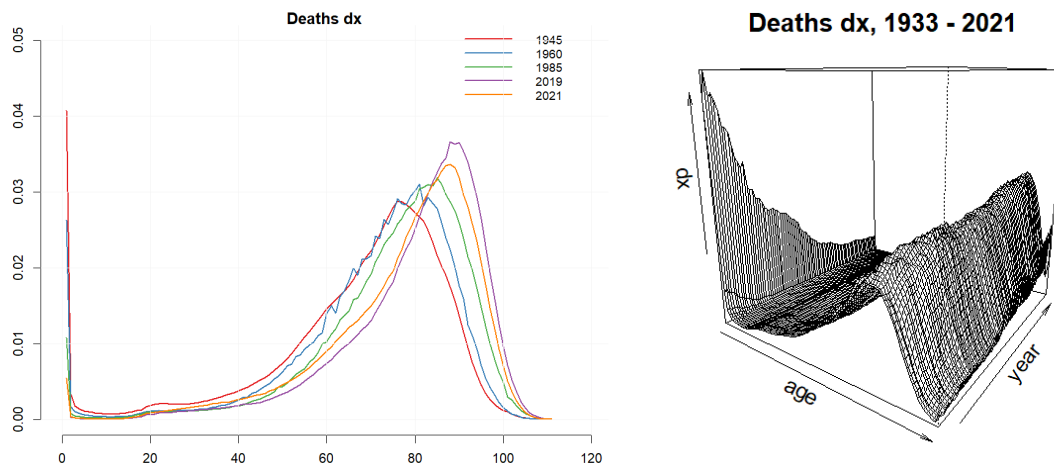


In the first years of the 1930s, there were fewer survivors at older ages compared to younger ages. This could be attributed to the lower life expectancy and higher mortality rates, especially among the elderly population.

From the mid-20th century the number of survivors at older ages started increasing, indicating improved longevity and quality of life. Then, as advancements in healthcare and living conditions kept happening, the number of survivors increased across all age groups.

The number of deaths (dx) represents the individuals who have died at a specific age and provides insights into mortality patterns and the factors influencing mortality rates over time. Figure 9 presents the evolution of the number of deaths in the United States in the period under study.

Figure 9. Evolution of the number of deaths from 1933 to 2021.



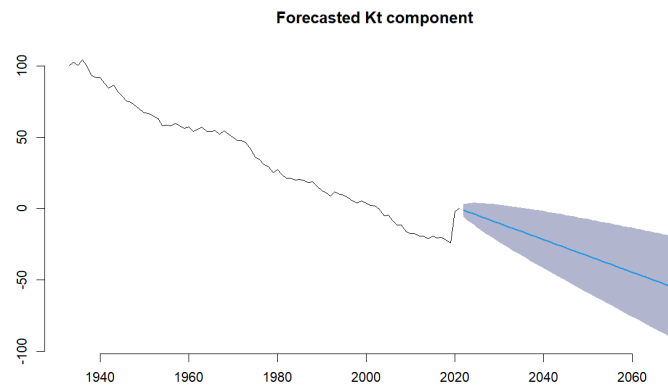
Until the end of World War II, the number of deaths was relatively high across all age groups, due to the limited access to healthcare and a high prevalence of infectious diseases. However, as the 20th century progressed, the number of deaths declined significantly. Advancements in medical knowledge, disease prevention, and public health interventions played a crucial role in reducing mortality rates. The implementation of widespread vaccination programs, improvements in sanitation, and increased availability of effective treatments contributed to the decline in the number of deaths across all age groups. Furthermore, targeted efforts to address specific health concerns, such as cardiovascular diseases and cancer, have resulted in substantial decreases in mortality rates.

The analysis of the number of survivors (lx) and deaths (dx) from 1933 to 2021 highlighted a significant improvement in mortality rates and survivorship experienced by the population. The advancements in medical treatments and public health measures have contributed to increased longevity and reduced mortality across all age groups.

The Lee-Carter Model

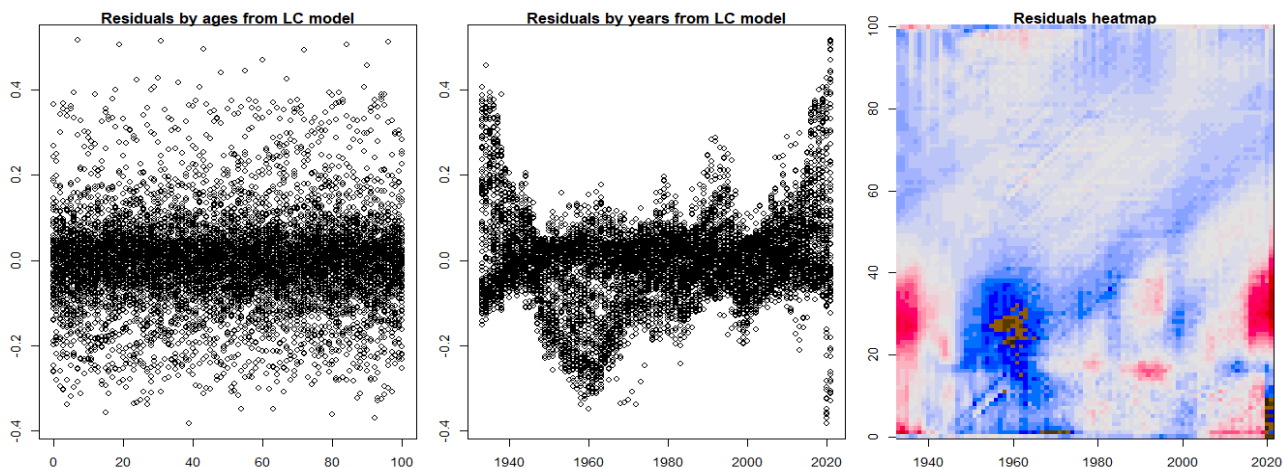
Starting from the number of survivors and the number of deaths at each age from 1933 to 2021, we estimated the parameters of the Lee-Carter model and used them to forecast life expectancy trend in the next 50 years with the lca function in R. The model managed to explain 95.5% of the variation in the data, meaning that it was able to capture almost all the underlying trend. Then we forecasted the $k(t)$ component, represented by the decreasing trend in Figure 10, which indicates a predicted decline in the mortality index joined by an improvement in survival rates.

Figure 10. Forecasted Kt component of the Lee-Carter model.



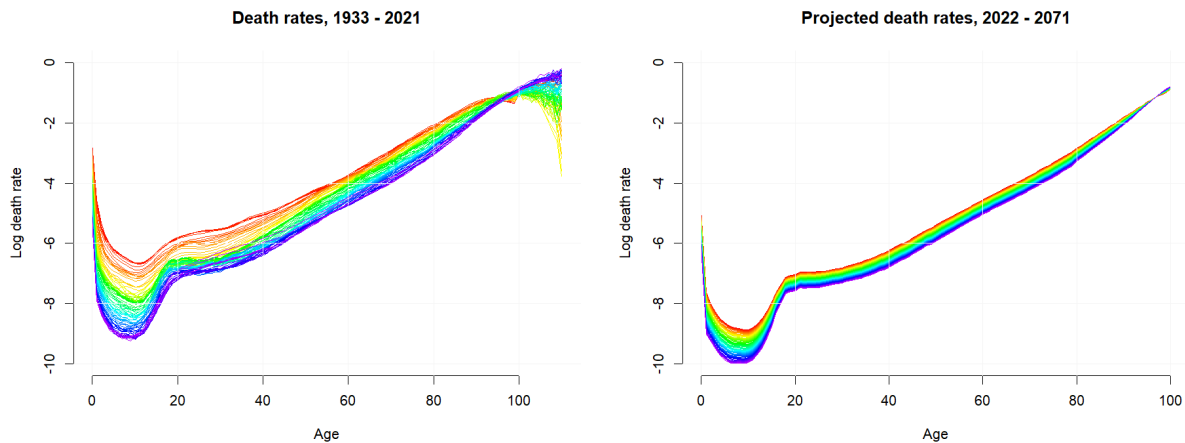
The next step was to conduct an analysis of the residuals to evaluate the discrepancies between the observed data and the predictions of the Lee-Carter at each age and year, which has been reported in Figure 11. From the residuals by year and the heatmap, there is evidence of a slight cohort effect, specifically for cohorts born before 1945.

Figure 11. Residuals of the Lee-Carter model.



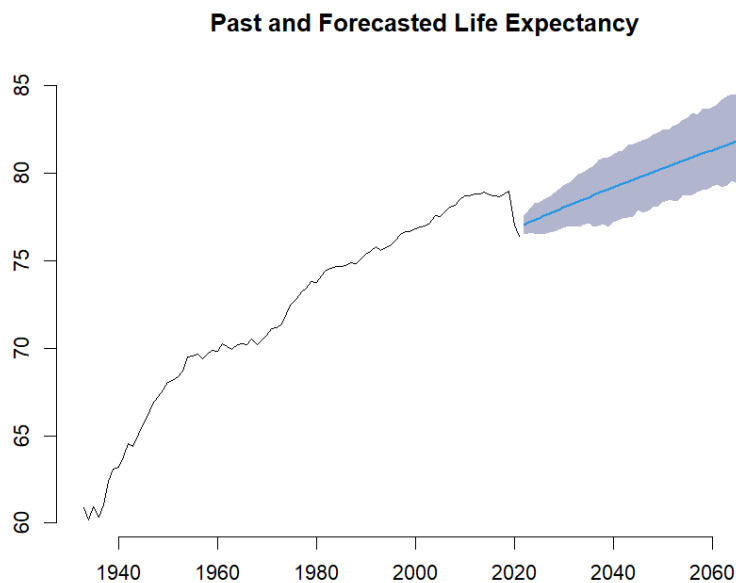
Then the lifetable function in R was employed to generate the projected lifetables based on the forecasted mortality rates. We forecasted the death rates and the result is reported in Figure 12, nearby the decreasing death rates observed from 1933 to 2021. The decreasing trend is projected to continue in the next 50 years across all ages, particularly in the first 30 years of life.

Figure 12. Death rates observed between 1933 (red) and 2021 (purple) and projected from 2022 (red) to 2071 (purple).



Through the projected death rates we were able to project life expectancy over the next 50 years, which is represented in Figure 13, showing an increasing trend which can be translated into improved longevity in the population.

Figure 13. Past and forecasted life expectancy at birth.



The Renshaw-Haberman model

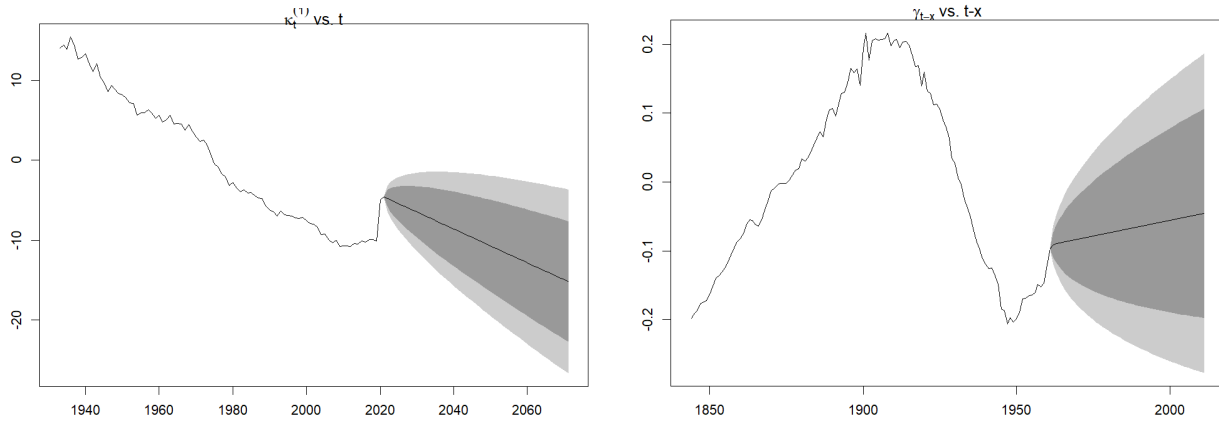
Since the heatmap of the residuals and their scatterplot by years showed some cohort effect, we also implemented the Renshaw-Haberman model, which may improve the model fitting as it also models the effect of the cohort of birth. In fact, the residuals of this model did not show any pattern anymore, as the effect of the year of birth was captured. The improvement of the model fitting was confirmed also by comparing the BIC values, which are reported in Table 1, showing that the Renshaw-Haberman model is the best one as it has the lowest value.

Table 1. BIC values for the Lee-Carter model and the Renshaw-Haberman model.

	LC model	RH model
BIC	96708.28	50423.10

Since this model gave a better fit to the data, we projected its parameters to understand the effect $k(t)$ and $\gamma(t, x)$, and we reported them in Figure 14. The projected $k(t)$ parameter is decreasing, confirming the declining trend in the mortality rate resulted from the Lee-Carter model, while the projected $\gamma(t, x)$ parameter shows a wide prediction interval due to the particular trend that was estimated in the past years.

Figure 14. Forecasted $k(t)$ and $\gamma(t, x)$ components of the Renshaw-Haberman model.

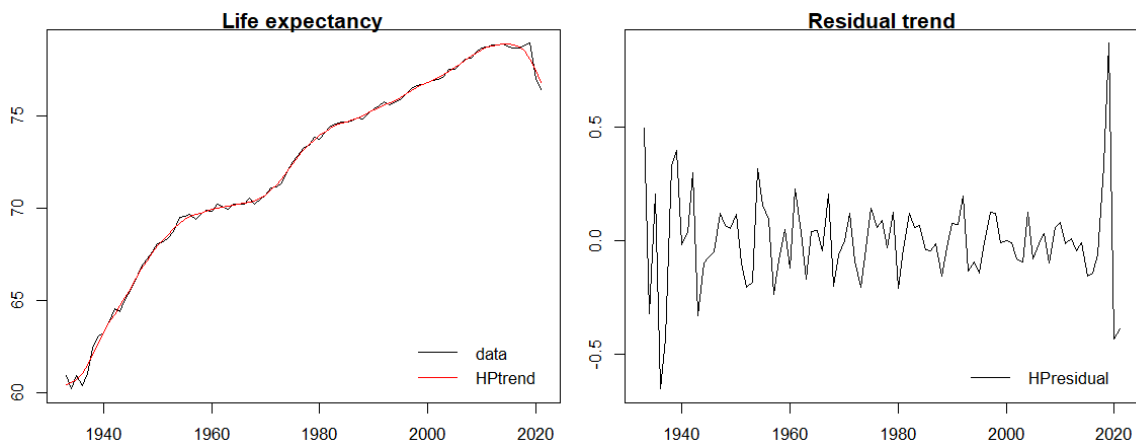


Detrend Life Expectancy

The next step was to perform a detrending analysis on the life expectancy data using two different filters: the Hodrick-Prescott (HP) filter and the Baxter-King (BK) filter. We recall that detrending is a technique used to separate the underlying trend or long-term component from the short-term fluctuations or residuals in a time series.

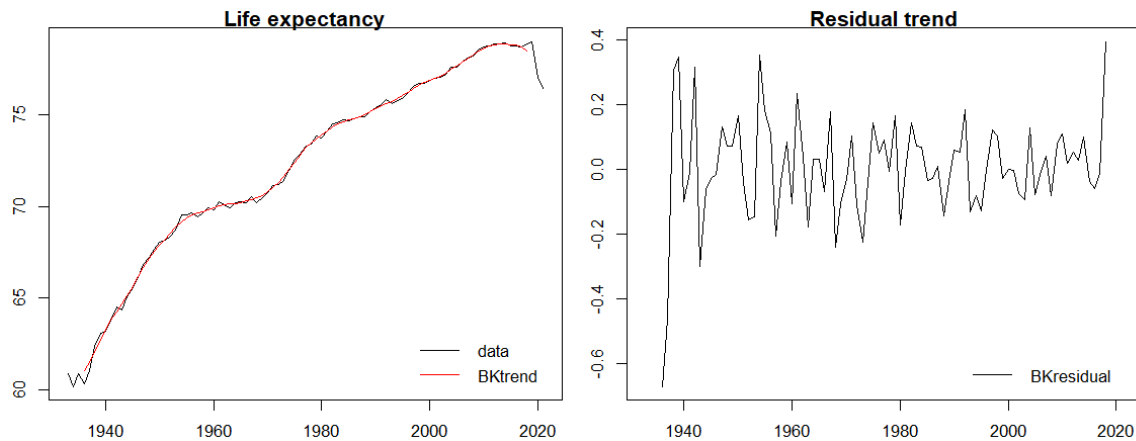
First, we applied the HP filter to the life expectancy data. This filter helps to smooth out the data and extract the trend component, which represents the long-term changes in life expectancy. The resulting trend component was then plotted alongside the original life expectancy data in Figure 15.

Figure 15. Life expectancy trend observed and smoothed with HP filter, and HP residual trend.



Next, we used the BK filter to detrend the life expectancy data. The BK filter also separates the trend component from the cyclical and irregular components in the time series. The filtered trend component was plotted, providing a visual representation of the long-term changes in life expectancy in Figure 16.

Figure 16. Life expectancy trend observed and smoothed with BK filter, and BK residual trend.



Conclusion

In this study, we explored how population structure and dynamics of the United States changed over the past century. The objective was to identify trends in mortality and fertility and to understand which factors had an influence and how was their effect. We aimed to provide insights into the demographic landscape of the United States and the implications that it has in various fields, including insurance, healthcare, pensions, and economic planning.

However, from our research it resulted that life expectancy in the United States has significantly increased over the period from 1933 to 2021 due to advancements in healthcare access, treatment availability, and disease prevention. However, fertility rates declined after the post-WWII demographic and economic expansion, reflecting changing societal attitudes towards family planning, and the prioritization of individual aspirations, especially for women.

Our findings have significant policy implications for several sectors. For instance, understanding the evolution of mortality rates and life expectancy is essential in the insurance industry to design appropriate risk models, but it is also important for the healthcare sector, which needs to adapt to the growing healthcare needs of an aging population. In addition, this results can be taken into account also for pension and economic planning.

Further research might investigate whether and how fertility and mortality trends are related to each other, considering different perspectives. For example, it could be of interest to explore the strength of the individual genetic contribution to longevity and to fertility to understand how much these phenomena are due to environmental external factors, and how much they depend on our genes. In addition, the inclusion of migration data and socioeconomic and cultural factors would provide a deeper understanding of these population dynamics.

In conclusion, the research highlighted the complex and dynamic nature of the American population and provided insights into the historical demographic landscape of the United States, trying to predict the trends for the next years. Indeed, by understanding these dynamics, policymakers are able to make more informed decisions to ensure the sustainability and well-being of the present and future population.

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